

CHAPTER V

Superstructures

As already defined superstructure is the part of structure above ground level which is visible easily. In a building columns, walls, beams, slabs, doors, window etc. form the superstructure while in bridges piers and deck form superstructure. The visible part of dams and water tank are superstructures of dam and water tank respectively. In this chapter various loads acting on superstructure are discussed and types of construction discussed.

5.1 TYPES OF LOADS ON SUPERSTRUCTURES

To get safe structures at the same time without ignoring economy of the structure, it is necessary to estimate the various loads acting suitably. Indian standard code IS: 875-1987 specifies various design loads for buildings and structures. They have grouped various loads as under:

1. Dead loads
2. Imposed loads
3. Wind loads
4. Snow loads
5. Earthquake loads
6. Special loads

Details of earthquake load is covered in IS: 1893 - 1984 which should be considered along with other types of loads given in IS-875. The code also gives various load combinations to be considered in the design.

5.2 DEAD LOADS

The dead load in a building comprises the weight of roofs, floors, beams, columns, walls, partition walls etc. which form permanent part of the building. It is to be found by working out volume of each part and then multiplying with unit weight. Unit weight of various materials are listed in part-I of IS: 875. Unit weights of some of the common materials are presented in Table 5.1.

Sl. No.

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TABLE 5.1. Unit Weight of Common Building Materials

Sl. No.	Material	Weight
1.	Brick Masonry	18.8 kN/m ³
2.	Stone Masonry	20.4 – 26.5 kN/m ³
3.	Plain cement concrete	24.0 kN/m ³
4.	Reinforced cement concrete	25.0 kN/m ³
5.	Timber	5 to 8 kN/m ³
6.	Wooden floors hard wood	160 kN/m ³
7.	Terrazo paving	200 N/m ³
8.	Country Tiles (single) including battens	240 N/m ³
9.	Mangalore Tiles with battens	700 N/m ³
10.	Mangalore tiles with flat tiles	650 N/m ³
11.	A.C. Sheets	785 N/m ³
12.	A.C. Sheet	170 N/m ³
	16 mm thick	110 N/m ³
	28 mm thick	
	6 mm thick	
	5 mm plain	

5.3 IMPOSED LOADS (IL)

The loads which keep on changing from time to time are called as imposed loads. Common examples of such loads in a building are the weight of the persons, weights of movable partition, dust loads and weight of furnitures. These loads were formerly known as live loads. These loads are to be suitably assumed by the designer. It is one of the major load in the design. The minimum values to be assumed are given in IS 875 (part 2)–1987. It depends upon the intended use of the building. These values are presented for square metre of floor area. The code gives the values of loads for the following occupancy classification:

(i) Residential buildings—dwelling houses, hotels, hostels, boiler rooms and plant rooms, garages.

(ii) Educational buildings

(iii) Institutional buildings

(iv) Assembly buildings

(v) Business and office buildings

(vi) Mercantile buildings

(vii) Industrial buildings, and

(viii) Storage rooms.

The code gives uniformly distributed load as well as concentrated loads. The floors are to be investigated for both uniformly distributed and worst position of concentrated loads. The one which gives worst effect is to be considered for the design but both should not be considered to act simultaneously.

In a particular building, imposed load may change from room to room. For example in a hotel or a hostel building the loads specified are,

	udl	Concentrated load
(a) Living rooms and bed rooms	2 kN/m ²	1.8 kN
(b) Kitchen	3 kN/m ²	4.5 kN
(c) Dining rooms	4 kN/m ²	2.7 kN
(d) Office rooms	2.5 kN/m ²	2.7 kN
(e) Store rooms	5 kN/m ²	4.5 kN
(f) Rooms for indoor games	3 kN/m ²	1.8 kN
(g) Bath rooms and toilets	2 kN/m ²	–
(h) Corridors, passages, stair cases etc.		
and (i) Balconies	4 kN/m ²	1.5 kN concentrated at outer edge.

Some of the important values are presented in table 5.2 which are the minimum values and wherever necessary more than these values are to be assumed.

TABLE 5.2. Minimum Imposed Load to be Considered

Sl. No.	Occupancy	UDL Load	Concentrated
1.	Bath rooms and toilets in all types of building	2 kN/m ²	1.8 kN
2.	Living and bed rooms	2 kN/m ²	1.8 kN
3.	Office rooms in	2.5 kN/m ²	2.7 kN
	(i) Hostels, hotels, hospitals and business building with separate store	3 kN/m ²	4.5 kN
	(ii) In assembly buildings	2 kN/m ²	1.8 kN
4.	Kitchens in (i) Dwelling houses	3 kN/m ²	4.5 kN
	(ii) Hostels, hotels and hospitals	3 kN/m ²	4.5 kN
5.	Banking halls, class rooms, x-ray rooms, operation rooms	3 kN/m ²	2.7 kN
6.	Dining rooms in (i) educational buildings, institutional and mercantile buildings	4 kN/m ²	2.7 kN
	(ii) hostels and hotels	3 kN/m ²	4.5 kN
7.	Corridors, passages, stair cases in	4 kN/m ²	4.5 kN
	(i) Dwelling houses, hostels and hotels	5 kN/m ²	4.5 kN
	(ii) Educational institutional and assembly buildings		
	(iii) Mercantile buildings		
8.	Reading rooms in libraries	3 kN/m ²	4.5 kN
	(i) With separate storage	4 kN/m ²	4.5 kN
	(ii) Without separate storage		
9.	Assembly areas in assembly buildings	5 kN/m ²	..
	(i) With fixed seats	5 kN/m ²	3.6 kN
	(ii) Without fixed seats	5 kN/m ²	4.5 kN
10.	Store rooms in educational buildings		
11.	Store room in libraries	6 kN/m ² for a height of 2.24 + 2 kN/m ² for every 1 m additional height	4.5 kN

12.	Boiler rooms and plant rooms in (i) hostels, hotels, hospitals, mercantile and industrial buildings	5 kN/m ²	4.5 kN
	(ii) Assembly & storage buildings	7.5 kN/m ²	4.5 kN

Imposed loads to be considered on various roofs are presented in table 5.3

TABLE 5.3. Imposed Loads on Various Types of Roofs
(Table 2 of National Building Code - 1983)

Sl. No.	Type of Roof	Imposed Load Measured on Plan Area	Minimum Imposed Load Measured on Plan
(i)	Flat, sloping or curved roof with slopes up to and including 10 degrees (a) Access provided	1.5 kN/m ²	3.75 kN uniformly distributed over any span of one metre width of the roof slab and 9 kN uniformly distributed over the span of any beam or truss or wall.
	(b) Access not provided except for maintenance.	0.75 kN/m ²	1.9 kN uniformly distributed over any span of one metre width of the roof slab and 4.5 kN uniformly distributed over the span of any beam of truss or wall.
(ii)	Sloping roof with slope greater than 10 degrees.	For roof membrane sheet or purlins - 0.75 kN/m ² less 0.02 kN/m ² for every degree increase in slope over 10 degrees.	Subject to a minimum of 0.4 kN/m ²
(iii)	Curved roof with slope of line obtained by joining springing point to the crown with the horizontal, greater than 10 degrees.	$(0.75 - 0.52 \alpha^2)$ kN/m ² where $\alpha = h/l$ h = height of the highest point of the structure measured from its springing; and l = chord width of the roof if singly curved and shorter of the two sides if doubly curved. Alternatively, where structural analysis can be carried out for curved roofs of all slopes in a simple manner applying the laws of statistics, the curved roofs shall be divided into minimum 6 equal segments and for each segment imposed load shall be calculated appropriate of each segment as given in (i) and (ii)	Subject to a minimum of 0.4 kN/m ²

NOTE 1. The loads given above do not include loads due to snow, rain, dust collection, etc. The roof shall be designed for imposed loads given above or snow/rain load, whichever is greater.

NOTE 2. For special types of roofs with highly permeable and absorbent material, the contingency of roof material increasing in weight due to absorption of moisture shall be provided for.

However in a multi-storeyed buildings chances of full imposed loads acting simultaneously on all floors is very rare. Hence the code makes provision for reduction of loads in designing columns, load bearing walls, their supports and foundations as shown in table 5.4.

TABLE 5.4. Reductions in Imposed Loads on Floors in Design of Supporting Structural Elements

Number of Floors (including the roof) to be carried by Member Under Consideration	Reduction in Total Distributed Imposed Load in Per cent
1	0
2	10
3	20
4	30
5 to 10	40
Over 10	50

5.4 WIND LOADS

The force exerted by the horizontal component of wind is to be considered in the design of buildings. It depends upon the velocity of wind and shape and size of the building. Complete details of calculating wind load on structures are given in IS-875 (Part 3) -1987. Brief idea of these provisions are given below:

(i) Using colour code, basic wind pressure ' V_b ' is shown in a map of India. Designer can pickup the value of V_b depending upon the locality of the building.

(ii) To get the design wind velocity V_z the following expression shall be used:

$$V_z = k_1 k_2 k_3 V_b$$

Where

k_1 = Risk coefficient

k_2 = Coefficient based on terrain, height and structure size.

k_3 = Topography factor

(iii) The design wind pressure is given by

$$p_z = 0.6 V_z^2$$

where p_z is in N/m^2 at height Z and V_z is in m/sec . Up to a height of 30 m, the wind pressure is considered to act uniformly. Above 30 m height, the wind pressure increases.

5.5 SNOW LOADS

IS 875 (part 4) - 1987 deals with snow loads on roofs of the building. For the building to be located in the regions wherever snow is likely to fall, this load is to be considered. The snow load acts vertically and may be expressed in kN/m^2 or N/m^2 . The minimum snow load on a roof area

or any other a expression

Where S

μ

S_0

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5.6 EARTHQUAKE

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The intensity of earthquake structure

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(a)

(b)

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or any other area above ground which is subjected to snow accumulation is obtained by the expression

$$S = \mu S_0$$

Where S = Design snow load on plan area of roof.

μ = Shape coefficient, and

S_0 = Ground snow load.

Ground snow load at any place depends on the critical combination of the maximum depth of undisturbed aggregate cumulative snow fall and its average density. These values for different regions may be obtained from Snow and Avalanches Study Establishment Manali (HP) or Indian Meteorological Department Pune. The shape coefficient depends on the shape of roofs and for some of the common shapes the code gives these coefficients. When the slope of roof is more than 60° this load is not considered.

It may be noted that roofs should be designed for the actual load due to snow or for the imposed load, whichever is more severe.

5.6 EARTHQUAKE FORCES

Earthquake shocks cause movement of foundation of structures. Due to inertia additional forces develop on super structure. The total vibration caused by earthquake may be resolved into three mutually perpendicular directions, usually taken as vertical and two horizontal directions. The movement in vertical direction do not cause forces in superstructure to any significant extent. But movement in horizontal directions cause considerable forces.

The intensity of vibration of ground expected at any location depends upon the magnitude of earthquake, the depth of focus, the distance from the epicenter and the strata on which the structure stands.

The response of the structure to the ground vibration is a function of the nature of foundation soil, size and mode of construction and the duration and intensity of ground motion. IS: 1893-1984 gives the details of such calculations for structures standing on soils which will not considerably settle or slide appreciably due to earthquake. The seismic accelerations for the design may be arrived at from **seismic coefficients**, which is defined as the ratio of acceleration due to earthquake and acceleration due to gravity. For the purpose of determining the seismic forces, India is divided into five zones. Depending on the problem, one of the following two methods may be used for computing the seismic forces:

- (a) Seismic coefficient method
- (b) Response spectrum method

The details of these methods are presented in IS 1893 code and also in National Building Code of India. After the Gujarat earthquake (2000) Government of India has realized the importance of structural designs based on considering seismic forces and has initiated training of the teachers of technical institution on a large scale (NPTEE).

There are large number of cases of less importance and relatively small structures for which no analysis be made for earthquake forces provided certain simple precautions are taken in the

construction. For example

- (a) Providing bracings in the vertical panels of steel and R.C.C. frames.
- (b) Avoiding mud and rubble masonry and going for light materials and well braced timber framed structures.

5.7 OTHER FORCES AND EFFECTS

As per the clause 19.6 of IS 456 – 2000, in addition to above load discussed, account shall be taken of the following forces and effects if they are liable to affect materially the safety and serviceability of the structure:

- (a) Foundation movement (See IS 1904)
- (b) Elastic axial shortening
- (c) Soil and fluid pressure (See IS 875, Part 5)
- (d) Vibration
- (e) Fatigue
- (f) Impact (See IS 875, Part 5)
- (g) Erection loads (See IS 875, Part 2) and
- (h) Stress concentration effect due to point load and the like.

5.8 LOAD COMBINATIONS

A judicious combination of the loads is necessary to ensure the required safety and economy in the design keeping in view the probability of

- (a) Their acting together
- (b) Their disposition in relation to other loads and severity of stresses or deformations caused by the combination of various loads.

The Recommended Load Combinations by National Building Codes

1.	DL	7.	DL + IL + EL
2.	DL + IL	8.	DL + IL + TL
3.	DL + WL	9.	DL + WL + TL
4.	DL + EL	10.	DL + EL + TL
5.	DL + TL	11.	DL + IL + WL + TL
6.	DL + IL + WL	12.	DL + IL + EL + TL

where

DL = dead load IL = imposed load

WL = wind load EL = earthquake load

and

TL = temperature load.

NOTE: When snow load is present on roofs, replace imposed load by snow load for the purpose of above load combinations.

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5.9 TYPES OF CONSTRUCTION

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Walls are an important part of superstructure. They are commonly constructed with stones, bricks or hollow concrete blocks. Walls enclose and divide the space in the building. In addition to it if they are made to carry load from roof/floor apart from self weight it is called **load bearing construction**.

If reinforced cement concrete or steel frame consisting of columns, beams, slabs are built first and walls are built only to enclose the area, the load transfer is mainly by beams and columns walls carry only self weight. These walls serve as filler material. Such structures are called **framed structures**.

Load bearing construction

Load bearing walls are built with stone, brick or concrete blocks joined together by cement mortar of 1 cement to 6 sand (1: 6). The walls are built course by course. Height of a course in stone (100 mm) respectively. In load bearing walls verticality of wall should be strictly ensured and vertical joints should be broken. The thickness of wall should be sufficient to transfer the load safely, without exceeding permissible stress. The critical portions in masonry from consideration of stresses are near the openings for doors and windows and the portion where concrete beams rest.

Minimum thicknesses used are 375 mm, 200 mm and 200 mm in case of stone, brick and hollow block constructions respectively. It is also recommended that slenderness ratio of wall defined as ratio of effective length or effective height to thickness should not be more than 27. National building code of India (NBC - 1983) defines effective height and effective length as given in table 5.5 and 5.6 [for full details refer NBC - 1983.]

TABLE 5.5. Effective Height of Walls in Terms of Actual Height H

Sl. No.	End Condition	Effective Height
1.	Lateral as well as rotational restraint	0.75 H
2.	Lateral as well as rotational restraint at one end and only lateral restraint at other	0.85 H
3.	Lateral restraint but no rotational restraint at both ends	1.0 H
4.	Lateral and rotational restraint at one end and no restraint at other ends (Compound walls, parapets etc.)	1.5 H

TABLE 5.6. Effective Length of Walls of Length L

Sl. No.	Condition of support	Effective Length
1.	Continuous and supported by cross walls	0.8 L
2.	Continuous at one end and supported by cross wall at other end	0.9 L
3.	Wall supported by cross walls of each end	1.0 L
4.	Free at one end and continuous at other end	1.5 L
5.	Free at one end and supported by cross wall at other end	2.0 L

Comparison between stone masonry and brick masonry

The merits and demerits of stone masonry and brick masonry are compared in table 5.7.

TABLE 5.7. Merits and Demerits of Stone Masonry and Brick Masonry

Description	Stone Masonry	Brick Masonry
1. Strength	High	Much Less
2. Durability	Excellent	Less
3. Appearance	Beautiful. No treatment is necessary with age.	Not so good. Needs plastering and colour washing.
4. Danger from dampness	No danger	May disintegrate.
5. Skill	Skilled Labour required for dressing and placing stones.	Ordinary skill is enough.
6. Handling	Heavy. Hence handling cost is more.	Easy to handle. Hence handling cost is less.
7. Fire resistance	Less	More.
8. Moulding to desired shape	Needs skilled labour	Convenient
9. Uses	For foundations, walls in building, dams, piers and abutment.	For building load bearing and position walls.

Framed constructions

Framed construction starts with foundations for columns. Columns are then raised. Beams and floors are built simultaneously in case of R.C.C. Construction goes floor by floor. After skeleton of second floors are ready construction of walls is taken up. Construction of multistorey buildings are possible in this type of construction.

Advantage of framed construction is interior alteration of rooms is possible by removing or by constructing additional walls.

In factories steel frame structures are also used. In these cases flooring is by R.C.C. and roofing is usually with trusses supporting A.C. sheets.

5.9 COMPOSITE CONSTRUCTION

If facing and backing of walls are made using different materials it is called composite wall construction. Facing material used is always good in appearance.

The following types of composite constructions are used:

1. Stone slabs facing with brick masonry backing.
2. Dressed stone facing and brick masonry backing.
3. Brick facing with rubble stone masonry.
4. Tile facing and brick backing.
5. Brick facing and concrete backing.
6. Stone facing and concrete backing.

In all these constructions proper bond between facing and backing should be achieved. For this purpose GI or aluminium clamps may be used. In case of brick facing alternate courses of bricks are projected inside backing. Rich plaster is used between facing and backing material.